BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

PHYSICS

Past Paper 2024 • Paper Code: PR–X (A-I-B4) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR–X (A-I-B4) (24)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. Time taken by oscillating object to complete one cycle is called:

(A) Frequency
(B) Time Period
(C) Amplitude
(D) Wavelength

Q2. When an object oscillate with a frequency of 0.5 Hz, its time period is:

(A) 1s
(B) 2s
(C) 5s
(D) 10s

Q3. In compression, the density of medium, as compared to surrounding is:

(A) Higher
(B) Lower
(C) Same
(D) Zero

Q4. When a ray of light enters from water to air, it moves:

(A) Towards the normal
(B) Away from normal
(C) Along the normal
(D) Un deviated

Q5. The speed of light in typical oil having refractive index 1.5 is:

(A) $2x10^8 m/s$
(B) $3x10^8 m/s$
(C) $4.5x10^8 m/s$
(D) $6x10^8 m/s$

Q6. When two capacitors of 15μF and 20μF are connected in series, the total capacitance is:

(A) 0.12μF (B) 8.5μF

(C) 20μF (D) 35μF

Q7. Resistance of a metallic wire is "R", if the length of wire is halved, its resistance becomes;

(A) R/2 (B) 2R

(C) R/4 (D) 4R

Q8. Induced emf in a coil can be increased by:

(A) Decreasing number of turns (B) Increasing magnetic field

(C) Decreasing length of coil (D) Increasing temperature of coil

Q9. The Boolean equation for NOR gate is:

(A) X=AB (B) X=A+B

(C) $X=\overline{A}+\overline{B}$ (D) $X=\overline{A}\,\overline{B}$

Q10. The applications and programming instructions that tell computer what to do are called:

(A) Software (B) Hardware

(C) Procedure (D) Information

Q11. The number of neutrons in a nucleus of $^{238}_{92}U$ is:

(A) 92 (B) 146

(C) 238 (D) 585

Q12. Complete the equation: $^{210}_{44}Po \rightarrow {}^{206}_{82}Pb$ + ...

(A) α (B) β^+

(C) β^- (D) γ

SECTION-B

Time Allowed: 2 Hours 45 Minutes

> Note: Note: Time allowed for Section-B and Section-C is 2 Hours and 45 minutes.

Q2. Answer any EIGHT parts. Each part carries FOUR marks.

i) Define loudness and list three factors affecting the loudness of sound.

ii) Define echo. In which medium air or water, an echo is heard sooner? Why?

iii) Prove through diagram θi = θr (Where θi = angle of incidence and θr = angle of reflection)

iv) Explain v = w/q and define volt.

v) State and explain Ohm's law.

vi) The current of 5A flows through a resistor of 7Ω for 3 minutes. Calculate the heat energy produced in resistor.

vii) Define A.C generator and describe the role of coil in the operation of A.C generator.

viii) Define CRO and describe the function of deflection system in CRC.

ix) Explain how a fax machine transmits documents over long distances.

x) Write symbol, composition, charge and effect on parent nucleus of α particle.

xi) Describe nuclear fission. Which particle is produced during fission?

SECTION-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 16

Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) Explain the conditions for a simple pendulum to execute Simple Harmonic Motion.

(b) Determine the magnitude of the restoring force on a pendulum bob of mass 125g, which has been displaced to an angel of 10° from the vertical. (g=9.8 m/s^2)

Q4.

(a) Discuss the role of lenses in the correctness of near-sightedness and far-sightedness.

(b) A biologist with a near-point distance N=25 cm examines an insect through a lens whose focal length is 4.4cm. Find the angular magnification when image produced at (a) near point (b) infinity.

Q5.

(a) State and explain Coulomb's law.

(b) Two point charges q1=7.8 μC and q2=3.5 μC are placed at a distance of 10cm. Find the Coulomb's force acting on each charge.

Q6.

(a) Explain right hand rule for finding the direction of force experienced by current carrying conductor in magnetic field also, derive the formula for magnetic force.

(b) A 0.8m long wire carries a current of 6.5 A, at right angle to uniform magnetic field of 0.07 T. Determine the force exerted on the wire.